Pavan Kumar Ambekar

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Professional Summary

Senior DFT Engineer with 6+ years of experience in advanced-node ASIC development (2nm–28nm), specializing in scan architecture, ATPG, MBIST, IJTAG, and RTL-level testability improvements. Proven track record of independently managing turn-key DFT projects end-to-end, from RTL assessment to ATPG sign-off. Adept at debugging complex DFT issues, optimizing coverage, and driving full DFT implementation across multi-million-gate designs. Leverages AI/LLM tools to automate repetitive DFT workflows (ATPG regression, DRC cleanup, coverage analysis), improving efficiency and turnaround time.

Experience

Kinara (NXP) (via Quest-Global) *Nov 2025 – Present*
Senior DFT Engineer — 5nm — Penang, Malaysia
Current Project

- Driving turn-key DFT delivery with RTL-level design modifications to enable testability.
- Implemented EDT and OCC insertion, optimizing compression and clock control strategies.
- Led scan insertion ensuring timing- and congestion-aware implementation.
- Developed and executed ATPG (Stuck-at and Transition Faults) with coverage improvement.
- Debugged ATPG coverage loss due to uncontrolled (UC) nets/flops and incorrect OCC placement.
- Performed RTL and Gate-Level Simulation (GLS) for validation, and formal equivalence checks (LEC).
- Collaborated with RTL and PD teams to resolve testability and timing issues.

Socionext, Japan (via Quest-Global) *Jun 2025 – Nov 2025*
Senior DFT Engineer — 5nm — Penang, Malaysia
Project: SHN-SNI

- Independently drove turn-key DFT delivery: owned end-to-end flow from RTL validation to ATPG sign-off.
- Performed SpyGlass DFT checks to clean RTL-level violations.
- Achieved 99.6% Stuck-at and 96% At-speed coverage.
- Executed scan insertion, ATPG, and simulation (zero-delay and timing).
- Conducted DFT sign-off checks.

MediaTek (via Mirafra Technologies) *Oct 2023 – Jun 2025*
Senior DFT & Synthesis Engineer — 2nm — Bangalore, India
Projects: Youshan, Davinci, Monet

- Performed synthesis, scan insertion, EDT/OCC insertion, and ATPG.
- Achieved 99.6% SA and 96% TDF coverage.
- Executed DFT validation, simulation, and formal checks (LEC, MUSE, CLP, SDC).
- Collaborated with cross-functional teams for coverage and integration closure.

Mirafra Technologies (Direct) *Jun 2023 – Sep 2023*
Senior DFT Engineer — 5nm — Bangalore, India

- Managed DFT process end-to-end for the Ariena Chip (defence project): MBIST insertion, synthesis, scan insertion, compression, ATPG.

CDAC, Trivandrum (via TechMahindra Cerium Systems) *Mar 2023 – Jun 2023*
DFT Engineer — 14nm — Bangalore, India
Project: Vega Quad-core Chip

- Owned turn-key DFT delivery for the Vega Quad-core Chip from MBIST insertion to pattern validation.
- Conducted MBIST and ICLNetwork tests with IJTAG graybox generation.
- Utilized custom PDL flow for testing ICLNetwork and IP from SoC level.

GUC, Taiwan (via TechMahindra Cerium Systems) *Nov 2022 – Feb 2023*
DFT Engineer — 7nm — Bangalore, India
- Validated DFT flow for Agamotto project based on in-house WFM data system.

Intel, US (via TechMahindra Cerium Systems) *Jun 2021 – Oct 2022*
DFT Engineer — 7nm — Bangalore, India
Project: Xeon Server Chip (Willow-Wood)
- Performed MBIST and ICLNetwork tests, generation of IJTAG graybox.
- Generated and tested ICLNetwork and MBIST from SoC level.
- Used custom PDL flow for testing ICLNetwork and IP.
- Conducted SDF / timing simulation.

TechMahindra (via Ziotron Consulting) *Dec 2020 – Apr 2022*
DFT Engineer — 7nm — Bangalore, India
- Handled 4 blocks: scan insertion, DRC cleaning, ATPG for Stuck-at and At-speed.
- Achieved 99.1% Stuck-at and 89% At-speed coverage.
- Performed coverage analysis and zero-delay simulation.

Technical Skills

- **Turn-key DFT Delivery:** Proven capability to independently own and deliver turn-key DFT projects from scratch (RTL assessment, scan/EDT/MBIST insertion, ATPG, coverage closure, simulation, sign-off) with minimal supervision
- **DFT and ASIC Flow:** Scan insertion, EDT/OCC compression, ATPG (SAF/TDF), MBIST, IJTAG, SpyGlass DRC, Synthesis, LEC, Timing/No-timing Simulation, Coverage analysis
- **AI/LLM Integration:** Leveraging AI/LLM tools to automate repetitive DFT workflows (ATPG regression, DRC cleanup, coverage analysis, script generation), improving productivity and reducing manual effort
- **EDA Tools:** Siemens Tessent, Cadence Genus, Synopsys DC, Formal Compiler, VCS, QuestaSim, Verdi, DVE
- **Programming:** Verilog, VHDL, PDL, TCL, Bash, Shell, HTML

Awards and Recognition

- **On-the-Fly Award** – Quest Global (2025), for exemplary project delivery and contributions.

Education

Master of Science (Microelectronics) *2020*
Newcastle University, Newcastle Upon Tyne, United Kingdom – First Class with Merit

Bachelor of Engineering (Electronics & Communication) *2018*
Siddaganga Institute of Technology (VTU), Tumkur, India – First Class